MILITARY SPECIFICATION

TOOL, POWER DRIVE TLU-437/E

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and inspection of the power drive tool, TLU-437/E.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-45208	Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MIL-STD-1332	Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of the DoD Mobile Electric Power Engine Generator Set Family
MIL-STD-45662	Calibration Systems Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 93), Lakehurst, NJ 08733, by using the self-addressed Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications forma part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1167AS100 Tool, Power Drive, TLU-437/E

1167AS103 Drive Tool Extension Assy

1167AS142 Cable Assy, Power Supply

DL1167AS100 Tool, Power Drive, TLU-437/E Data List

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4.1 and 6.3).

3.2 Parts and material. Unless otherwise specified in the contract or purchase order (see 6.2.1h), material and parts utilized in the fabrication of the tool, part number (p/n) 1167AS100-1, shall conform to the drawings, specifications, and requirements listed in DL1167AS100 or as specified herein. There shall be no deviations from the prescribed material, parts or processes without prior approval of the aquiring activity.

3.2.1 Recycled virgin and reclaimed materials. There is no exclusion to the use of recycled or reclaimed materials and no mandate for the use of virgin materials provided it meets the requirements of this specification.

3.3 Design and construction. The tool shall be of the design, construction, and physical dimensions specified on drawings listed in DL1167AS100 or as specified herein (see 6.2.11). There shall be no deviations from dimensions or tolerances without prior approval of the acquiring activity.

3.4 Lubrication. Lubrication shall be applied to parts and surfaces as specified on drawings listed in DL1167AS100 in such a manner that it provides long life and smooth operation of the tool.

3.5 Input power. The tool shall be connectable to 120 volts (V), single phase, 2 wire, 400 hertz (Hz), class 2C, alternating current (at) as defined by MIL-STD-1332.

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3.6 Logistics support program. A logistics support program shall be prepared and conducted in accordance with attachment number 1, Integrated Logistic Support Detail Specification for Power Drive Tool TLU-437/E.

3.7 Operation. Interfacing surfaces of the tool output shaft, Power Pack Adapter, p/n A51S69315-15, Gun Systems Input Shaft, p/n BA226, (see 3.11) and Drive Tool Extension Assy, p/m 1167AS103-1, shall be capable of mating. The tool shall operate smoothly in a no load condition without binding or chattering when tested as specified in 4.6.1.

3.8 Performance. With torque limiter set to maximum capacity, unless otherwise specified, and with the power cable connected to the input power source (see 3.5), the tool shall meet the performance parameters listed in table I when tested as specified in 4.6.2.

TABLE I. Performance parameters

Minimum torque (ft-lb)1/	Minimum speed (rpm)1/	Maximum current (A)1/
20	385	22
30	365	28
40	340	34
48	320	39

1/ ft-lb....foot pounds
rpm.....revolutions per minute
A.....Amperes

3.9 Finish. The tool shall have protective coating, surface texture, and finish as specified on the drawings listed in DL1167AS100.

3.10 Identification marking. The tool shall be marked as specified on the drawings listed in DL1167AS100 or in accordance with MIL-STD-130 if not specified on the drawings.

3.11 Government furnished equipment. Items of government furnished equipment (GFE) required are A51S69315-15 Power Pack Adapter and BA226 Gun System Input Shaft (see 6.4). These items are used to determine drive tool Interface compatibility and to verify operation at no load.

3.12 Workmanship. Parts shall be free from chips, dirt, grit or other foreign material. All material shall be free from cracks, flaws and any other injurious defects which may impair the serviceability of the tool.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth In the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. An inspection system program (see 6.2.2) shall be prepared and provisions to provide and maintain an inspection system in accordance with MIL-I-45208 shall be made.

4.1.2 Test instruments. All test instruments shall be calibrated in accordance with MIL-STD-45662.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article Inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all Inspections shall be performed under standard ambient conditions which are:

a. Temperature: 73° ± 18°F (23° ± 10°C)

b. Relative humidity: 50% ± 30%

c. Atmosphere pressure: 28.5 + 2.0 inches of mercury (hg)
- 3.0

(725 + 50 millimeter (mm) of hg)
- 75

4.4 First article inspection. First article inspection shall be performed on sample units produced with equipment and procedures normally used in production (see 6.2.1e and 6.3). Sample units shall be subjected to the inspections listed in tables II and III. Failure to pass any of the inspections listed in tables II and III shall be cause for rejection (see 6.2.1n).

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TABLE II. First article inspection and classification of defect.

Classification of defect	Examination or test	Requirement paragraph	Method paragraph
Major			
	Material	3.2	4.6.4
	Motor	3.2	4.6.4
	Motor protection	3.2	4.6.4
	Power supply cable and connector	3.2	4.6.4
	Torque limiter	3.2	4.6.4
	Switches	3.2	4.6.4
	Output interface	3.3	4.6.4
	Protective coating	3.9	4.6.4
	Bearings	3.2	4.6.4
	Lubrication	3.4	4.6.4
Minor			
	Identification missing, incorrect or inadequate	3.10	4.6.4

TABLE III. First article inspection.

Examination or test 1/	Requirement paragraph	Method paragraph
Operation	3.7	4.6.1, 4.6.1.1
Performance	3.8	4.6.2, 4.6.2.1

1/ Perform examination or test in the order listed.

4.4.1 Sample. Unless otherwise specified in the contract or purchase order, the first article sample shall consist of two complete tools. The approved first article sample units shall be completed within the timeframe specified (see 6.2.1j. 6.2.1o, and 6.3).

4.4.2 Test report. A test report (see 6.2.1m and 6.2.2) shall be prepared within the timeframe specified (see 6.2.1k). The report shall include the following:

a. Test equipment and procedures.

b. All test data obtained from applicable tests.

c. Description of malfunctions, damages, failures and adjustments (other than adjustments permitted by this specification) which occur during inspection.

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d. Causes and analysis of malfunctions, damages or failures and reasons for adjustments.

e. Corrective action taken or required.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections conducted on each tool as listed in tables IV and V. Failure to pass any of the inspections listed in tables IV and V, except for the minor defect (identification) listed in table IV, shall be cause for rejection of the tool.

TABLE IV. Quality conformance inspection and classification of defect.

Classification of defect	Examination or test	Requirement paragraph	Method paragraph
Major			
	Power supply cable and connector	3.2	4.6.4
	Output interface	3.3	4.6.4
	Protective coating	3.9	4.6.4
Minor			
	Identification missing, incorrect or inadequate	3.10	4.6.4

TABLE V. Quality conformance inspection.

Examination or test 1/	Requirement paragraph	Method paragraph
Operation	3.7	4.6.1, 4.6.1.1
Performance	3.8	4.6.2, 4.6.2.2

1/ Perform examination or test in the order listed.

4.5.1 Incident report. A report (see 6.2.1m and 6.2.2) shall be prepared within 24 hours of any incident of equipment malfunction or failure during conduction of the inspections listed on tables IV and V and shall be available to the acquiring activity. Included in the report shall be the date of the incident and method of detection.

4.5.2 Failure report. A report (see 6.2.1m and 6.2.2) shall be prepared within 15 days of a failure Incident and shall be available to the acquiring activity.

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4.6 Methods of inspection.

4.6.1 Operation. The tool and ancillary components shall be tested in accordance with test method 1 (see 4.6.1.1). Defectives are those failing to meet the requirements of 3.7.

4.6.1.1 Test method 1. Test method 1 shall consist of the following test apparatus and test procedure:

a. Test apparatus

1. Power Pack Adapter (GFE), p/n A51S69315-15.

2. Electric power supply in accordance with 3.5.

3. Drive Tool Extension Assy, p/n 1167AS103-1.

4. Gun System Input Shaft (GFE), p/n BA226.

b. Test procedure

1. Connect the power pack adapter (GFE) to the tool output drive adapter and the tool output drive shaft. Check for binding when connecting, locking, and unlocking the power pack adapter.

2. Run the tool, with the power pack adapter connected, for 30 seconds in each direction. Check the tool for binding or chattering. Remove the power pack adapter from the tool.

3. Install the drive tool extension assy to the tool output drive shaft and then dismantle. Check any unusual force exerted in installing and dismantling.

4. Insert the drive tool extension to the gun system input shaft (GFE) and then disconnect. Check for binding when inserting and removing the drive tool extension.

4.6.2 Performance. The tool output performance shall be tested in accordance with test method 2A (see 4.6.2.1) for first article inspection and test method 2B (see 4.6.2.2) for quality conformance inspection. Torque and speed shall be measured at the tool output shaft. Defective are those failing to meet the requirements of 3.8.

4.6.2.1 Test method 2A (first article). Test method 2A shall consist of the following test apparatus and test procedure:

a. Test apparatus

1. Suggested test stand (see 4.6.2.3) or prony brake or power absorbing dynamometer (with minimum torque rating at 75 ft-lb and maximum speed rating over 500 rpm and with accessories to adapt tool outut shaft).

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2. Tachometer, or other suitable rotational speed measuring device.

3. Electric power supply, in accordance with 3.5, equipped with voltage and amperage recording devices.

b. Test procedure

1. Secure the tool to prevent inadvertent movement and position the tachometer to obtain output shaft rotational speed.

2. Using the power cable, p/n 1167AS142-1, connect the tool to a 120 Vac, 400 H_z source.

3. Using the control switches on the tool, perform a no-load speed test for 4 minutes in each direction. Record the output shaft rotational speed for each shaft direction. Record the voltage and amperage at output of the power supply.

4. Attach the tool to the prony brake or the dynamometer with sufficient support and alignment to minimize wobble and resist torque reaction to the tool.

5. Run the tool in the clockwise (CW) direction at each incremental torque load specified below. Allow 2 minutes dwelltime between each incremental run. Record the torque, speed, voltage and amperage draw. Run time for each incremental load is as follows:

Load (ft-lb)	Run time (minutes)
20	4
30	4
40	2
48	1

6. Repeat step 5 three times. Record the torque, speed, voltage and amperage draw for each run.

7. Repeat steps 5 and 6 except run the tool in the counterclockwise (CCW) direction.

8. Test the torque limiter by adjusting to 20 ft-lb and installing the tool as in step 4.

9. Run the tool in the clockwise direction with a 15 ft-lb load, then, rapidly increase the torque load to activate the torque limiter. Continue to run the tool for 5 seconds with the torque limiter activated. Record the torque limiter actuation torque, voltage and amperage.

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10. Repeat step 9 three times. Record the torque limiter actuation torque, voltage and amperage draw for each run.

11. Repeat steps 9 and 10 except run the tool in the counterclockwise direction.

12. Increase the torque limiter to a 50 ft-lb capacity and the dynamometer a to 40 ft-lb load. Perform steps 9, 10 and 11.

4.6.2.2 Test method 2B (quality conformance). Test method 2B shall consist of the following test appartus and test procedure:

a. Test apparatus

1. Suggested test stand (see 4.6.2.3) or prony brake or power absorbing dynamometer (with minimum torque rating at 75 ft-lb and maximum speed rating over 500 rpm and with accessories to adapt tool output shaft).

2. Tachometer, or other suitable rotational speed measuring device.

3. Electric power supply, in accordance with 3.5, equipped with voltage and amperage recording devices.

b. Test procedure

1. Secure the tool to prevent inadvertent movement and position the tachometer to obtain output shaft rotational speed.

2. Using the power cable, p/n 1167AS142-1, connect the tool to a 120 Vat, 400 Hz source.

3. Using the control switches on the tool, perform a no load speed test for 4 minutes in each direction. Record the output shaft rotational speed for each shaft direction. Record the voltage and amperage at output of the power supply.

4. Attach the tool to the prony brake or the dynamometer with sufficient support and alignment to minimize wobble and resist torque reaction to the tool.

5. Run the tool in the clockwise direction at each incremental torque load specified below. Allow 2 minutes dwelltime between each incremental run. Record the torque, speed, voltage and amperage draw. Run time for each incremental load is as follows:

Load (ft-lb)	Run Time (minutes)
20	4
30	4
40	2
48	1

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6. Repeat step 5 except run the tool in the counterclockwise direction.

7. Test the torque limiter by adjusting to 20 ft-lb and installing the tool as in step 4.

8. Run the tool in the clockwise direction with a 15 ft-lb load, then, rapidly increase the torque load to activate the torque limiter. Continue to run the tool for five seconds with the torque limiter activated. Record the torque limiter actuation torque, voltage and amperage draw.

9. Repeat step 8 except run the tool in the counterclockwise direction.

10. Increase the torque limiter to a 50 ft-lb capacity and the dynamometer to a 40 ft-lb load. Perform steps 8 and 9.

4.6.2.3 Test stand and fixtures. The suggested test stand, as shown in figure 1, supports the unit in its normal orientation at its normal support position. A dummy output shaft is coaxial to a pneumatic activated disk prony brake. The prony brake is trundled on self aligning ball bearings. A moment arm fastened to the prony brake housing is coupled to a force scale that is reached at the test stand base. The force scale reads pounds in either direction. The specified torque level requirements are met by regulating the pressure to the prony brake. Torque is measured by multiplying the length of the moment arm by the pounds of force recorded on the scale. Output shaft speed is monitored by a magnetic speed pick-up feeding off of a 60 tooth gear pressed on the unit output shaft. Voltage and current are monitored by dedicated meters.

4.6.3 Examination. Tools shall be inspected for conformance to the requirements of this specification and DL1167AS100, with respect to parts, material, design, construction, lubrication, input power, finish, marking, and workmanship (see 3.2, 3.3, 3.4, 3.5, 3.9, 3.10 and 3.12 respectively).

4.7 Inspection of packaging. The sampling and inspection of the preservation-packaging, packing and container marking shall conform to this specification (see section 5).

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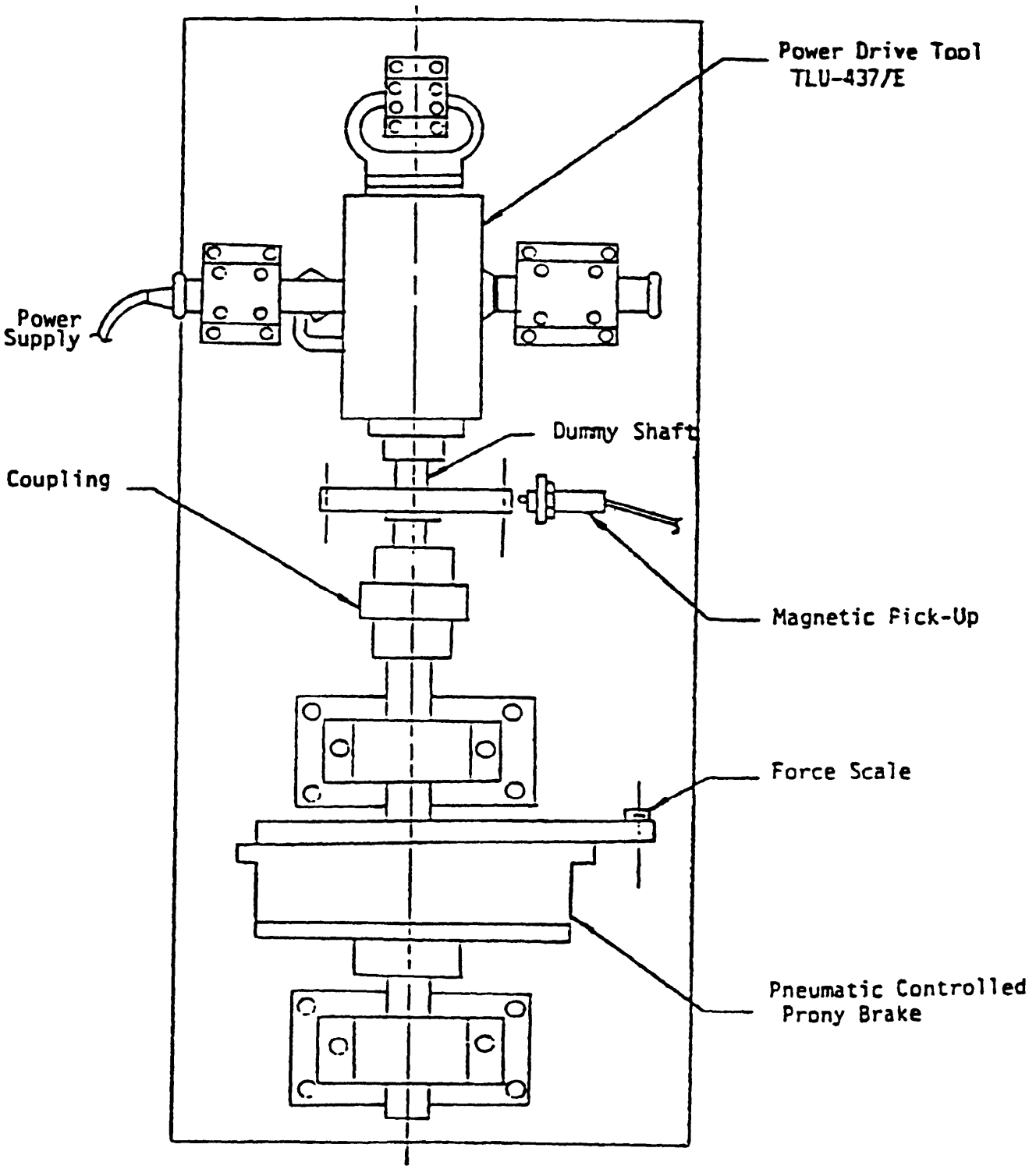


FIGURE 1. Suggested test stand.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-STD-794, level A or C as specified (see 6.2.1d).

5.2 Packing. Packing shall be in accordance with MIL-STD-794, level A or C as specified (see 6.2.1d).

5.3 Unitized loading. Unitized loading is authorized in accordance with MIL-STD-794.

5.4 Marking. In addition to any special or other identification marking required by the contract or purchase order (see 6.2.1d), shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The power drive tool is intended for use as a primary power source for aircraft ammunition and external stores loading aboard ship and shore stations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition requirements should specify the following:

a. Title, number and date of this specification.

b. Part number

c. Quantity required.

d. Level of preservation-packaging, packing and marking.

e. Whether first article is required (see 6.3).

f. First article sample acceptance, if other than specified in section 4 of this specification.

g. Production lot acceptance, if other than specified in section 4 of this specification.

h. Parts and material, if other than specified in 3.2.

i. Design and construction, if other than specified in 3.3.

j. Timeframe for submission of first article sample units.

k. Timeframe for submission of first article test report.

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l. Name and location of Government approved test laboratory.

m. Where test reports and data items (see 6.2.2) are to be forwarded.

n. Responsibility for disassembling, reassembling, and retesting.

o. Whether first article sample units should be refurbished and considered as part of the contract or purchase order.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL Incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
a. 4.1.1	Inspection Systems Program Plan	DI-R-4803	----
b. 4.4.2	Acceptance Test Reports	DI-T-3721A	----
c. 4.5.1	Report, Failure/Malfunction	UDI-T-23724B	----
d. 4.5.2	Report, Failure	UDI-R-21141A	----

(Data item description related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. When a contractor is in continuous production of power drive tools from contract to contract, consideration should be given to waive the first article inspection. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents, and disposition of the first article.

6.4 Government furnished equipment. The contracting officer should arrange to furnish the equipment listed in 3.11. GFE should be requested from the Naval Air Engineering Center, Support Equipment Engineering Department, Code 92721, Lakehurst, NJ 08733.

Preparing activity:
Navy - AS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85703(AS)	TOOL, POWER DRIVE TLU-437/E

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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